# Swirl Jet Diffuser VS



## Description
Swirl jet diffuser with fixed and radial arranged blades.

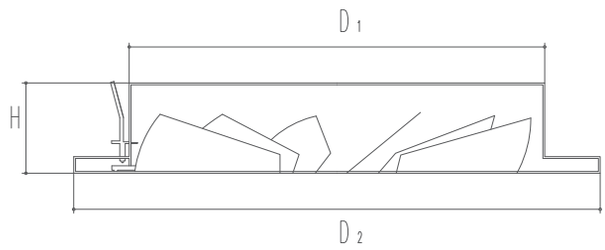
## Application
Supply/ exhaust of the air.

## Characteristics
Recommended for rooms of 2.60 to 4.00 m in height.
The design, radial layout and vortex action of the blades leads to a high level of induction rate.
Can be used in systems with both constant or variable air volumes.

## Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

## Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016
Other RAL colors available on request

## Technical design




| Ø | D1 | H | D2 |
|---|---|---|---|
| 125 | 123 | 50 | 158 |
| 160 | 158 | 53 | 197 |
| 200 | 198 | 53 | 241 |
| 250 | 248 | 50 | 295 |
| 315 | 313 | 53 | 364 |
| 400 | 398 | 53 | 450 |

## Selection table VS

| Air flow (m³/h) | Ø Ak [m²] | 125 0.0025 |
|---|---|---|
| 35 | Veff(m/s) | 3.9 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 8.2 |
| 43 | Veff(m/s) | 4.8 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 13.5 |
| 50 | Veff(m/s) | 5.6 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 18.0 |
| 61 | Veff(m/s) | 6.8 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 28.0 |
| 76 | Veff(m/s) | 8.4 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 39.0 |
| 96 | Veff(m/s) | 10.7 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 60.0 |

| Air flow (m³/h) | Ø Ak [m²] | 160 0.004 |
|---|---|---|
| 43 | Veff(m/s) | 3.0 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 7.5 |
| 52 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 9.0 |
| 65 | Veff(m/s) | 4.5 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 15.0 |
| 81 | Veff(m/s) | 5.6 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 23.0 |
| 100 | Veff(m/s) | 6.9 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 35.0 |
| 120 | Veff(m/s) | 8.3 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 50.0 |

| Air flow (m³/h) | Ø Ak [m²] | 200 0.007 |
|---|---|---|
| 90 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 12.0 |
| 120 | Veff(m/s) | 4.8 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 18.0 |
| 140 | Veff(m/s) | 5.6 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 25.0 |
| 160 | Veff(m/s) | 6.3 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 35.0 |
| 185 | Veff(m/s) | 7.3 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 48.0 |
| 230 | Veff(m/s) | 9.1 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 70.0 |

| Air flow (m³/h) | Ø Ak [m²] | 250 0.011 |
|---|---|---|
| 190 | Veff(m/s) | 4.8 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 15.0 |
| 240 | Veff(m/s) | 6.1 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 20.0 |
| 280 | Veff(m/s) | 7.1 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 30.0 |
| 320 | Veff(m/s) | 8.1 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 40.0 |
| 410 | Veff(m/s) | 10.4 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 58.0 |
| 520 | Veff(m/s) | 13.1 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 60.0 |

## Selection table VS

| Air flow (m³/h) | Ø / Ak [m²] | 315 / 0.021 |
|---|---|---|
| 240 | Veff(m/s) | 3.2 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 9.0 |
| 270 | Veff(m/s) | 3.6 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 13.5 |
| 320 | Veff(m/s) | 4.2 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 18.0 |
| 380 | Veff(m/s) | 5.0 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 25.0 |
| 470 | Veff(m/s) | 6.2 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 35.0 |
| 530 | Veff(m/s) | 7.0 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 48.0 |

| Air flow (m³/h) | Ø / Ak [m²] | 400 / 0.028 |
|---|---|---|
| 290 | Veff(m/s) | 2.9 |
| | NR [dB(A)] | 17.0 |
| | △ Pt [pa] | 10.0 |
| 330 | Veff(m/s) | 3.3 |
| | NR [dB(A)] | 22.0 |
| | △ Pt [pa] | 15.0 |
| 350 | Veff(m/s) | 3.5 |
| | NR [dB(A)] | 27.0 |
| | △ Pt [pa] | 20.0 |
| 500 | Veff(m/s) | 5.0 |
| | NR [dB(A)] | 32.0 |
| | △ Pt [pa] | 30.0 |
| 600 | Veff(m/s) | 6.0 |
| | NR [dB(A)] | 37.0 |
| | △ Pt [pa] | 42.0 |
| 720 | Veff(m/s) | 7.1 |
| | NR [dB(A)] | 42.0 |
| | △ Pt [pa] | 65.0 |

## Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside diffuser
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

## Order codes

| Type | + | ADP IZ/N | / | Ø | / | 0 |
|---|---|---|---|---|---|---|
| **VS** | | Insulated/ Non-insulated adapter | | 125<br>160<br>200<br>250<br>315<br>400 | | **0** Painted white RAL<br>**R** Other RAL paints |